PULSE WIDTH MODULATION OF LIGHT EMITTING DIODES IN A SPREAD SPECTRUM CONTROLLED SYSTEM AND ANALYSIS IN OCTAVE

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Abstract— This work describes the design, implementation, and analysis of pulse width modulation of light emitting diodes in a spread spectrum-controlled system and analysis in octave, an open-source numerical calculation software. This paper focuses on the creation of an efficient pulse width modulation, a technique that can be used for precise control and eliminating flickering in LEDs. The suggested architecture starts with signal creation, modulation, noise addition, then demodulation procedure using Fast Fourier Transform (FFT) operations. Simulations and performance evaluations are used to assess the system's Bit Error Rate (BER) and Signal-to-Noise Ratio (SNR). The results show that pulse width modulation is suitable and effective in improving the efficiency of LED lighting systems. This paper is divided into different sections; Section I gives brief Introduction of this work, section II discusses about the materials and methods, section III presents the results and discussion, and section IV explains the conclusion.

Keywords—*LEDs, Flickering, Frequency, Pulse width modulation, spread spectrum, dimming.*

I. INTRODUCTION

Light Emitting Diode is a semiconductor device that produces light when electric current passes through it. The light-emitting diode (LED) is today's most rapidly developing lighting technology compared with other types of lighting, LEDs are energy efficient, environmentally friendly, smaller in size, cheaper, emit low heat, durable and last longer. This makes LEDs a preferred choice for various lighting and display applications from indicator lights and displays to decorative, general lighting, automotive lighting, and even in areas like horticulture and food preservation..

However, LEDs lighting system encounter issues such as poor dimming and flickering which occurs due to rapid fluctuations in light output, which can impair output performance. LED flickering, often imperceptible to the naked eye, can lead to discomfort, eye strain, and even health issues.

PWM dimming is widely used in LED drivers due to its simplicity and high efficiency. This paper introduces a synchronized spread spectrum control (SSC) method integrated with PWM dimming. The proposed method ensures that the converter's switching frequency at the start of each PWM pulse remains consistent, even while the frequency varies within the pulse. This alignment produces a predictable current waveform, effectively eliminating perceptible flickering.

II. MATERIALS AND METHODS

A. Material

Software: Octave is an open-source numerical calculation software used to construct and simulate PWM signals. Octave has two toolboxes, the signal processing toolbox and the communications toolbox, which are required for different signal processing functions and communication methods.

Hardware:

LEDs, DC-DC converters, resistors, MOSFETs etc.

Computer Hardware: A typical personal PC for running Octave and doing simulations.

B. Methods

Design

This block diagram illustrates a Spread Spectrum Controlled (SSC) PWM LED driver system. It is designed to reduce visible flicker in LED lighting, especially at low PWM dimming levels.

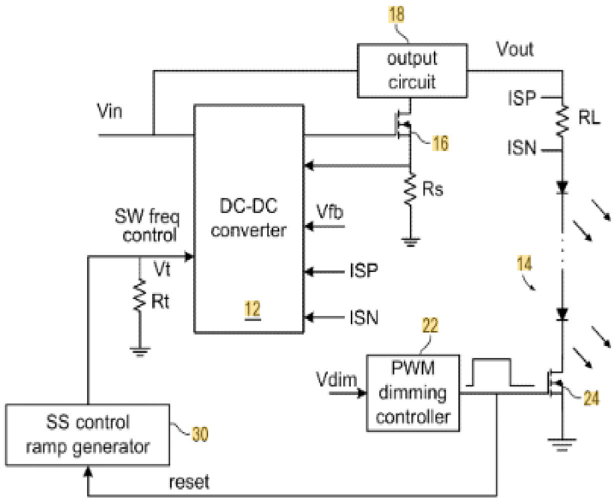

Fig. 1 The structure of SSC PWM LED driver system

The Structure of a Spread Spectrum Controlled PWM LED driver system is shown in Fig.1 above. The operational Principle is as follows;

1. DC-DC Converter (Block 12)

Converts an input voltage Vin to a regulated output voltage or current required for driving LEDs. Feedback signals ISP and ISN are used to regulate the current output via sensing resistor Rs.

2. Output Circuit (Block 18)
Manages LED load RL, supplies regulated voltage Vout to the LEDs and helps protect the converter output.

3. LED String (Block 14)
Represents multiple LEDs connected in series. The current through this LED string is the important parameter that is being regulated.

4. Switching Frequency Control
Controlled by resistor Rt and voltage Vt, which determine the switching frequency of the DC-DC converter.
These parameters can be dynamically altered by the SSC Control Ramp Generator to introduce frequency modulation.

5. SSC Control Ramp Generator (Block 30)
Generates a slowly varying signal (e.g., triangular or sine wave) that modulates the switching frequency.
This reduces the spectral concentration of EMI and synchronizes frequency variation across PWM cycles to prevent visible flicker. A reset signal aligns the SSC ramp to the start of each PWM pulse.

6. PWM Dimming Controller (Block 22)
Receives a dimming control voltage Vdim and outputs a PWM signal. This PWM signal drives the switch (24) in series with the LED string. Controls the on/off duration of LED current, effectively adjusting brightness.

Simulation
SSC-PWM Simulation Steps in Octave
A. Signal Generation: Creation of a triangular or sawtooth waveform to vary the PWM switching frequency (e.g., 300–500 kHz). It is used to modulate the timing of PWM pulses (frequency or duty cycle).

B. PWM Modelling: Defining high and low logic levels based on input binary data or control voltages. Modulates duty cycle (0–100%) corresponding to desired dimming level.

C. Frequency Modulation (SSC): Frequency dithering, spreading out the noise spectrum of the power supply.
The PWM frequency is not fixed but swept within a range using the SSC signal. Reduces EMI by spreading energy across a wider frequency band.

D. Time & Frequency Analysis: FFT analysis provides valuable insights into the frequency content of PWM/SSC signals, enabling effective frequency domain filtering for flicker elimination, and simulation of LED output current response to modulated PWM.

Analysis
Performance of Bit Error Rate (BER) analysis for the system was carried out, simulating the transmission and reception of binary data modulated with the combined SSC (Spread Spectrum Control) and PWM signal in Octave.
Bit Error Rate (BER) at various SNR levels were observed in order to evaluate system performance. Signal-to-Noise Ratio (SNR) shows how SNR affects system performance and efficiency under various noise circumstances.
Simulation Scenarios evaluate the system under a variety of scenarios and noise levels, to ensure a thorough analysis.

IMPLEMENTATION
This is the implementation of PWM in Octave. The code provides a waveform of a triangle-like SSC frequency waveform that sweeps between 300 kHz and 400 kHz. A 500 Hz PWM signal (50% duty) aligned with each sweep, ensuring repeatable timing.
Running the code below in Octave will offer visual representation of waveforms, as well as a performance evaluation using the BER vs. SNR plot.

Below are Octave simulation codes showing received signals at various switching frequencies under a Spread Spectrum Control (SSC) system with PWM dimming.
It demonstrates how the SSC-modulated switching frequency affects the shape of the received signal, which could represent LED current or brightness under PWM control.

OCTAVE COMPLETE CODE:
```
clear; clc;
% Simulation time and resolution
T_sim = 5e-3;              % 5 ms simulation
dt = 1e-6;              % Time step: 1 µs
t = 0:dt:T_sim;
% SSC parameters
f_base = 350e3;            % Base switching frequency
(Hz)
f_dev = 50e3;              % Deviation (±50 kHz)
f_pwm = 500;               % PWM frequency (used for
sync)
T_pwm = 1 / f_pwm;

samples_per_pwm = round(T_pwm / dt);
% Build SSC waveform (triangle pattern synchronized
to PWM period)
f_ssc = zeros(size(t));
for i = 0:floor(length(t)/samples_per_pwm) - 1
    start_idx = i * samples_per_pwm + 1;
    stop_idx = min(start_idx + samples_per_pwm - 1,
length(t));
    span = stop_idx - start_idx + 1;

    triangle = linspace(f_base + f_dev, f_base - f_dev,
span);  % Down-slope
```

```
  f_ssc(start_idx:stop_idx) = triangle;
end
% Plot SSC waveform
figure;
plot(t*1e3, f_ssc/1e3, 'r-', 'LineWidth', 1.5);  %
Convert Hz to kHz
xlabel('Time (ms)');
ylabel('Switching Frequency (kHz)');
title('Synchronized Spread Spectrum Control (SSC)
Waveform');
grid on;
% Add reference lines
yline(f_base/1e3, 'k:');                  % Base frequency
yline((f_base + f_dev)/1e3, 'g:');        % Max freq
yline((f_base - f_dev)/1e3, 'g:');        % Min freq
legend('SSC Frequency', 'Base Freq', 'Max Freq', 'Min
Freq');
```

```
clear; clc;
% Simulation time and resolution
T_sim = 5e-3;              % 5 milliseconds
dt = 1e-6;                 % Time step: 1 µs
t = 0:dt:T_sim;
% PWM parameters
f_pwm = 500;               % PWM frequency in Hz
T_pwm = 1 / f_pwm;         % PWM period
duty_cycle = 0.5;          % 50% duty cycle
% Generate PWM waveform
pwm_signal = mod(t, T_pwm) < (duty_cycle *
T_pwm);
% Plot PWM waveform
figure;
plot(t * 1e3, pwm_signal * 100, 'b', 'LineWidth', 1.5);
% Convert to percentage
xlabel('Time (ms)');
ylabel('PWM Signal (%)');
title('PWM Signal (500 Hz, 50% Duty Cycle)');
ylim([-10 110]);
grid on;
legend('PWM Output');
```

```
clear; clc;
% Simulation time and resolution
T_sim = 5e-3;              % 5 ms total simulation
dt = 1e-6;                 % Time step (1 µs)
t = 0:dt:T_sim;
% SSC and PWM parameters
f_pwm = 500;               % PWM frequency (Hz)
T_pwm = 1 / f_pwm;         % PWM period (s)
f_base = 350e3;            % Base switching frequency (Hz)
f_dev = 50e3;              % Frequency deviation (Hz)
samples_per_pwm = round(T_pwm / dt);
% Generate synchronized SSC waveform (triangle)
f_inst_sync = zeros(size(t));
for i = 0:floor(length(t)/samples_per_pwm) - 1
   start_idx = i * samples_per_pwm + 1;
   stop_idx = min(start_idx + samples_per_pwm - 1,
length(t));
   span = stop_idx - start_idx + 1;
   triangle_segment = linspace(f_base + f_dev, f_base -
f_dev, span);
   f_inst_sync(start_idx:stop_idx) = triangle_segment;
end
% Generate PWM signal (square wave)
pwm_signal = mod(t, T_pwm) < (T_pwm / 2);  % 50% duty
cycle
pwm_scaled = pwm_signal * (f_base + f_dev)/1e3;  % scale
for plotting
% ---------------------------------------------
% Plot SSC waveform and PWM signal on same axis
% ---------------------------------------------
figure;
plot(t*1e3, f_inst_sync/1e3, 'r-', 'LineWidth', 1.5); hold on;
plot(t*1e3, pwm_scaled, 'b--', 'LineWidth', 1.2);
xlabel('Time (ms)');
ylabel('SW Frequency (kHz) & Scaled PWM');
title('Synchronized SSC Waveform with PWM Pulse');
grid on;
```

% BER Calculation

```
clear; clc;
% Parameters
N_bits = 1e5;                   % Number of bits to transmit
EbN0_dB = 0:2:20;               % Eb/N0 range in dB
EbN0 = 10.^(EbN0_dB/10);        % Linear scale
ber = zeros(size(EbN0));        % Store BER values
% SSC PWM modulation parameters (conceptual)
f_pwm = 500;                    % PWM frequency (Hz)
ssc_sweep_range = 100e3;        % SSC sweep (e.g., 100
kHz)
symbol_duration = 1/f_pwm;      % Symbol duration =
PWM period
% Loop over each Eb/N0 value
for k = 1:length(EbN0)
   % Generate random bits
   bits = randi([0 1], 1, N_bits);
   % OOK Modulation (LED ON for 1, OFF for 0)
  tx_signal = bits;
   % Add Gaussian noise
   noise = sqrt(1/(2*EbN0(k))) * randn(1, N_bits);
   rx_signal = tx_signal + noise;
   % Demodlation (Threshold = 0.5)
   rx_bits = rx_signal > 0.5;
   % Calculate BER
   ber(k) = sum(rx_bits ~= bits) / N_bits;
end

% Plot BER vs. Eb/N0
figure;
semilogy(EbN0_dB, ber, 'bo-', 'LineWidth', 1.5,
'MarkerFaceColor', 'b');
xlabel('SNR)');
ylabel('Bit Error Rate (BER)');
```

```
title('BER vs. SNR for SSC-Compatible PWM Modulated
LED System');
grid on;
legend(' SSC+PWM');
```

III. RESULTS AND DISCUSSION

A. Results

The performance of Pulse Width Modulation system for the LEDs system was assessed using extensive simulations in Octave. The major measures studied were Bit Error Rate (BER) and Signal-to-Noise Ratio.

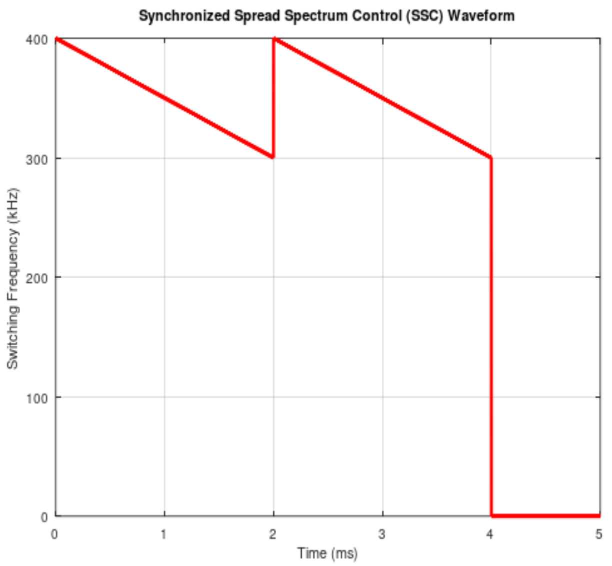

Fig.2 Plot showing the Synchronized Spread Spectrum Control (SSC) waveform

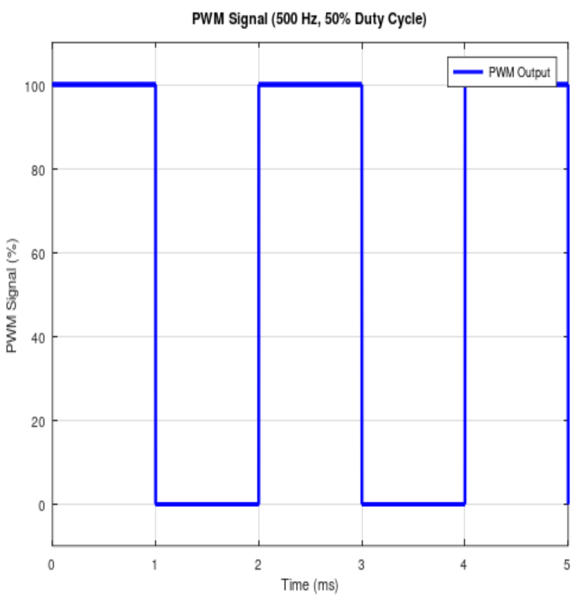

Fig.3 Plot showing the PWM Signal

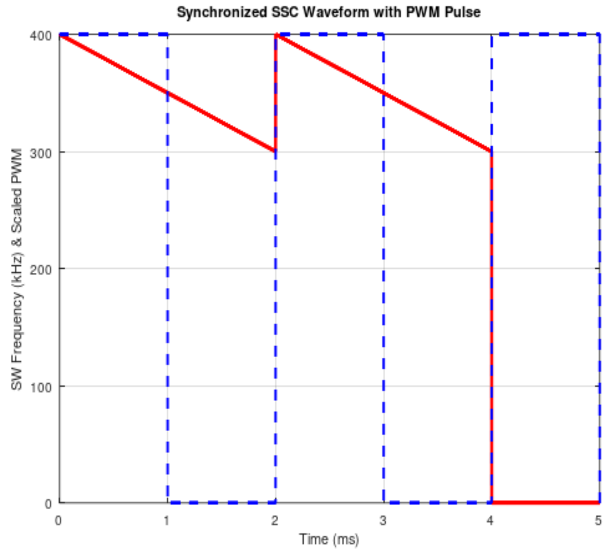

Fig.4 Plot showing the Synchronized Spread Spectrum Control (SSC) waveform and PWM pulse

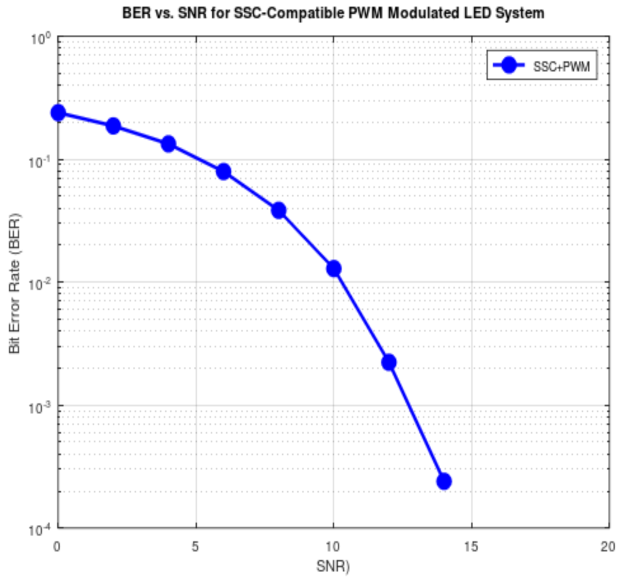

Fig. 5 BER VS SNR

B. Discussion

Fig 2 and 3 shows generated Synchronized Spread Spectrum Control (SSC) waveform and the PWM Signal respectively. Fig. 4 - Simulates a PWM signal (50% duty cycle, 500 Hz). Then multiplies the signal with a sinusoidal carrier representing different SSC switching frequencies (e.g., 300, 350, 400 kHz). SSC synchronized with PWM resets the SSC ramp at each PWM pulse, producing consistent current waveforms and eliminating flicker, which reduces noise sources that degrade SNR. The resulting received signals show how the SSC-controlled frequency modulates the PWM carrier. Dotted lines mark the base, min and max switching frequencies for easy reference. LED flicker occurs a result of the rapid switching of the LED's on and off states. While the frequency of this switching is typically high enough that some humans don't perceive it as flicker, some individuals are sensitive to it, leading to perceived flicker. To mitigate this, PWM technique is employed to smooth out the signal and eliminate the perceived flicker.

The findings demonstrate the efficacy of the Pulse Width Modulation method implemented in Octave for LEDs systems. The large reduction in BER with rising SNR demonstrates the system's capacity to offer reliable non flickering LED lighting across various applications.

SSC modulates the switching frequency to spread electromagnetic interference (EMI), reducing noise peaks and improving signal quality. PWM controls the LED brightness and data encoding. Synchronization between PWM and SSC ensures consistent current waveforms, reducing flicker and improving signal stability. Together, these reduce noise and distortions.

Bit Error Rate:
Bit Error Rate (BER) and Signal-to-Noise Ratio (SNR) are two important metrics for assessing the performance of digital communication systems, such as SSC-compatible PWM modulated LED systems.

The BER performance was measured at various SNR values ranging from 0 to 20db. The results from fig.5 showed that raising SNR resulted in a considerable reduction in BER, demonstrating that the system was resistant against noise. At a signal-to-noise ratio of 13 dB, the BER was around 10^-3, demonstrating the system's efficiency at high SNR circumstances.

Signal to Noise Ratio (SNR):
The effects of SNR on system performance were investigated as shown in fig. 5.
It was observed that the system performed effectively even at low SNR levels, with a substantial reduction in BER as SNR increases.

The use of Octave as the simulation platform is very helpful, offering a reliable environment for simulation and visual plots. Octave's open-source nature makes it easy to modify and visualize, supporting future study and advancement in this field.

IV. CONCLUSION

By employing spread spectrum control synchronized with the PWM dimming signal, the switching frequency of the LED driver converter is modulated in a way that maintains waveform consistency. This results in a perceptibly flicker-free LED operation, suitable for both low and high brightness levels. This procedure is practical for enhancing LED lighting quality across various fields of application. Future work might include analysis on how synchronized a spread spectrum controlled PWM LED driver system affects LED junction temperature and long-term reliability due to high-frequency switching.

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